chapter eight

CONFIDENCE INTERVAL ESTIMATION

When you can measure what you are speaking about
and express it in numbers you know something about it;
but when you cannot measure it,
when you cannot express it in numbers,
your knowledge is of a meager and unsatisfactory kind.
– Lord Kelvin

INTRODUCTION

In chapter 7, we used the Central Limit Theorem to answer probability questions about a sample statistic. The Central Limit Theorem guaranteed us that when $n \geq 30$, the sampling distribution was approximately normal. We converted values of $\bar{x}$ and $\hat{p}$ into z scores and used the standard normal distribution to answer questions about the normal distribution. Our work in chapter 7 set the stage for what is known as **inferential statistics**.

Inferential statistics involves two key ideas; **estimation** and hypothesis testing. In this chapter we focus on estimation, and in chapter 9 we will focus on hypothesis tests. Briefly stated, estimation involves using sample statistics, such as $\bar{x}$ and $\hat{p}$, to estimate the value of the related population parameters, such as μ and p.

For example, the sample mean $\bar{x}$ is what we call a **point estimate** of the population parameter μ. However, point estimates are not considered very useful because we do not have a way of deciding on the accuracy of our estimate of the population parameter.

There is a second form of estimation that includes both a statement of our confidence in the estimate and the **maximum error of the estimate**. This estimate involves an interval that is constructed around the point estimate. We call this estimate a **confidence interval**. The confidence interval is constructed from either a z or t score using the Central Limit Theorem.

For example with a large sample, we find the **upper and lower limits** of the confidence interval for μ by writing the z score for $\bar{x}$ and solving for the population parameter μ.

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \Rightarrow \mu = \left(\bar{x} \pm z \frac{s}{\sqrt{n}} \right)$$

The value of the z score is determined by the **level of confidence** we want for our interval. For example, if we want to be 90% confident in our estimate of μ, we use a z score that gives

$$\frac{100\% - 90\%}{2} = 5\%$$

of the area under the standard normal curve to the right of z. In the case of the 90% confidence interval for μ, we use $z = 1.65$. So we would say that we were 90% confident that μ would be in the interval

$$\left(\bar{x} - 1.65 \frac{s}{\sqrt{n}}, \bar{x} + 1.65 \frac{s}{\sqrt{n}} \right)$$

We finish the chapter by focusing on the size of the sample we need to limit the error of our estimate of the population parameter.

CHAPTER 8 HOT SPOTS

1. **The Difference Between A Point And An Interval Estimate**
   Starts on **page 8–4.** Problems on **pages 8–8, 8–9.** Mann 8.1,8.2

2. **Degrees Of Freedom.**
   Starts on **page 8–10.** Mann 8.4.1

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic of a help session with your professor.

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Mann Sections 8.1 And 8.2

Hot Spot #1 – The Difference Between A Point And An Interval Estimate

Earlier in the text we used the sample mean, $\bar{x}$, to estimate the population mean, μ. The sample mean is called a point estimate of the population parameter. We need to take a sample from the population to obtain the data we use to calculate the sample mean. If the size of the sample is large, we feel good about our estimate. On the other hand if the sample size is small, we may feel uneasy about our point estimate of the parameter. In either situation, we do not have any way of finding out how accurate we are in our estimate.

In contrast, the confidence interval gives us a way of estimating a population parameter that includes a measure of how confident we are and the maximum error of our estimate. Constructing a confidence interval involves the same ideas we used when we worked with the Empirical Rule with a known value of μ to find an interval that contained $\bar{x}$ 95% of the time. We simply reverse the process in finding a confidence interval. We start with a known value of $\bar{x}$, and find an interval that will contain the true value of μ 95% of the time.

We will first look at using the Empirical Rule with the results of the Central Limit Theorem to find an interval that contains the sample mean, $\bar{x}$, 95% of the time. In this case we know the value of μ. As an example let x be a continuous random variable that represents SAT scores for a normal population with

$$\mu = 1{,}200 \text{ and } \sigma = 100 \text{ and } n = 100.$$

The Empirical Rule tells us that 95% of the sample means should be in the interval

$$\mu \pm \frac{2\sigma}{\sqrt{n}}.$$

The interval includes sample means from $1{,}200 - \frac{2(100)}{\sqrt{100}}$ to $1{,}200 + \frac{2(100)}{\sqrt{100}}$

The interval is $(1{,}180\,,\,1{,}220)$ and is illustrated in the following graph:

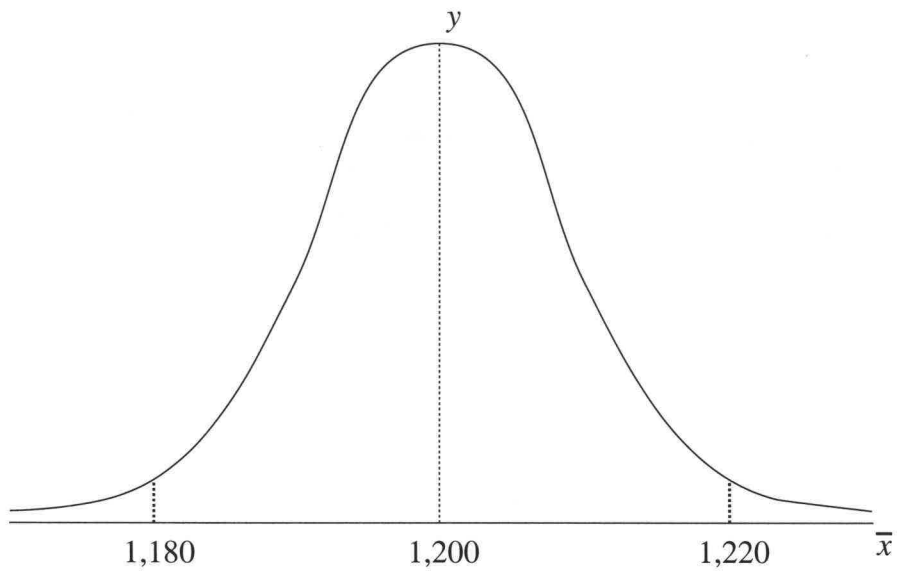

Figure 8–1: Interval from 1,180 to 1,220

In contrast, when we construct a confidence interval, we do not know the value of μ. So we find $\bar{x}$ and use the idea of a z score to create an interval that contains μ. We start with the z score:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

We want what we will call the 95% confidence interval for estimating the population mean, μ. We will be 95% confident that the interval we construct from $\bar{x}$ will contain the true value of μ. In this sense we will accept being wrong 5% of the time. The z score for the 95% confidence interval is found by getting a value for $z\left(\frac{0.05}{2}\right)$. When $z = 1.96$, we have 2.5% of the area under the normal curve in the right tail. So we will use $z \pm 1.96$ to get our 95% confidence interval.

We now substitute for z and rewrite the z score. Solving for μ we get the following:

$$\mu = \bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Generally speaking, we do not know σ. So, we use the sample standard deviation s as an estimate for σ. Since we are using the Central Limit Theorem to find $\sigma_{\bar{x}}$, it is important that the sample size be large, $n \geq 30$.

Returning to our example on SAT test scores, we will consider the case where we have no idea of the value of μ. As an illustration we will assume that all we have is a sample with

$$\bar{x} = 1{,}210 \text{ and } s = 100 \text{ and } n = 100.$$

We want to find the 95% confidence interval for μ. So we substitute the values of $\bar{x}$, s, and n into the formula as follows:

formula

- $\mu = \bar{x} \pm 1.96 \dfrac{\sigma}{\sqrt{n}}$

substituting

- $\mu = 1{,}210 \pm 1.96 \dfrac{100}{\sqrt{100}}$

simplifying

- $\mu = 1{,}210 \pm 19.6$

interval

- $1{,}190.4 \leq \mu \leq 1{,}229.6$

We then say that we are 95% confident that the interval from 1,190.4 to 1,229.6 contains the population mean μ.

We usually do not know if the interval actually contains the true value of μ. We can only say we are 95% confident that the interval contains μ. However, to see how the confidence interval works, we assumed we knew that $\mu = 1{,}200$. In the above example, the confidence interval we constructed did contain $\mu = 1{,}200$.

Now, let us see what it means to be 95% confident that the interval contains μ. We need to return to what we did with the Empirical Rule, where we used a value of 2 rather than the more accurate z score of 1.96. We found that if $\mu = 1{,}200$ then 95% of the sample means would be in the interval from 1,180 to 1,220. If we use any

$\bar{x}$ in this interval to construct the confidence interval for μ, the confidence interval will contain μ.

The 95% confidence interval for μ we constructed from $\bar{x} = 1{,}210$ contained the value of 1,200. We need to note that $\bar{x}$ was in the interval from 1,180 to 1,220 that we obtained from the Empirical Rule. In fact, if we construct the confidence interval using any $\bar{x}$ from the interval $(1{,}180\,,1{,}220)$, the confidence interval will contain the value 1,200.

Only 5% of the $\bar{x}$'s are outside of the interval from 1,180 to 1,220. If we use a $\bar{x}$ outside the interval $(1{,}180\,,1{,}220)$, the confidence interval we construct will not contain 1,200. To show this we will use $\bar{x} = 1{,}175$ and $s = 100$ and $n = 100$.

- $\mu = \bar{x} \pm 1.96\dfrac{\sigma}{\sqrt{n}}$ *formula*
- $\mu = 1{,}175 \pm 1.96\dfrac{100}{\sqrt{100}}$ *substituting*
- $\mu = 1{,}175 \pm 19.6$ *simplifying*
- $1{,}155.4 \le \mu \le 1{,}194.6$ *interval*

This time the 95% confidence interval goes from 1,155.4 to 1,194.6 and the confidence interval does **not** contain $\mu = 1{,}200$.

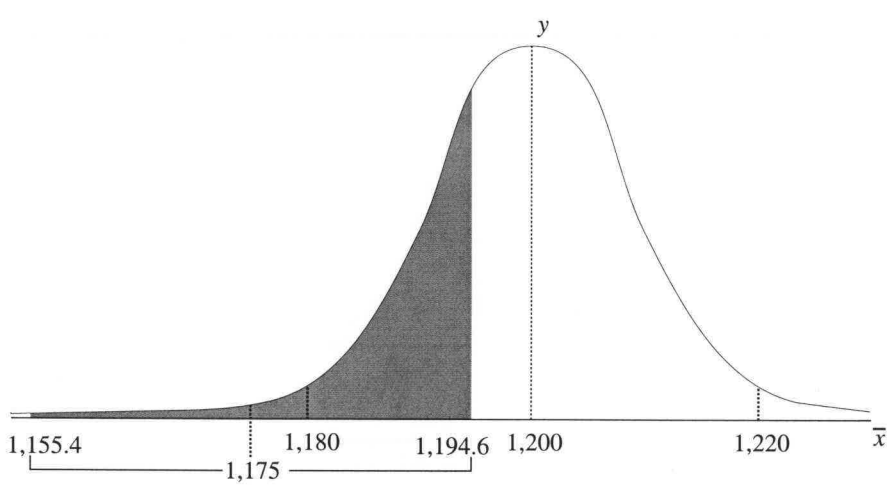

Figure 8–2: Illustration Of Choosing An $\bar{x}$ Outside The 95% Confidence Interval

The following table helps summarize the results of the examples:

Table 8–1: Summary Of Analysis Of Confidence Interval

Summary
1. $\bar{x}$ is in the interval $(1{,}180\,,1{,}220)$ 95% of the time.
2. We use an $\bar{x}$ in the interval to construct the confidence interval.
3. The confidence interval contains μ=1200.
1. $\bar{x}$ is **not** in the interval $(1{,}180\,,1{,}220)$ 5% of the time.
2. We use an $\bar{x}$ outside the interval to construct the confidence interval.
3. The confidence interval does **not** contain μ=1200.

Another way of saying what is in the table is that if:

1. We take 100 different random samples, with $n \geq 30$;
2. Find $\bar{x}$ and s for each sample;
3. Construct the confidence interval for each $\bar{x}$, then 95 of the 100 confidence intervals will contain $\mu = 1{,}200$.

Hot Spot #1 Sample Problem and Solution: Given the distribution of SAT scores discussed above with $\bar{x} = 1{,}210$, $s = 100$, and $n = 100$ show that the 95% confidence interval contains $\mu = 1{,}200$.

Solution: Step 1. Substitute values of $\bar{x}$, s, and n into $\mu = \bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$.

$$\mu = 1{,}210 \pm 1.96\frac{100}{\sqrt{100}}$$

Step 2. Determine the upper and lower values of the interval $\mu = 1{,}210 \pm 19.6$.

$1{,}190.4 \le \mu \le 1{,}229.6$

Step 3. The interval from 1,190.4 to 1,229.6 contains $\mu = 1{,}200$.

Sample Problem and Answer: Given the distribution of SAT scores discussed above with $\bar{x} = 1{,}250$, $s = 100$, and $n = 100$ show the 95% confidence interval does **not** contain $\mu = 1{,}200$.

Answer:

- $\mu = \bar{x} \pm 1.96 \dfrac{\sigma}{\sqrt{n}}$ *formula*
- $\mu = 1{,}250 \pm 1.96 \dfrac{100}{\sqrt{100}}$ *substituting*
- $\mu = 1{,}250 \pm 19.6$ *simplifying*
- $1{,}230.4 \le \mu \le 1{,}269.6$ *interval*

The confidence interval does **not** contain $\mu = 1{,}200$.

Sample Problem and Answer: Given the following five values of $\bar{x}$ for the SAT scores discussed above, a 95% confidence interval is to be constructed for each $\bar{x}$. Which one of the values of $\bar{x}$ will result in the confidence interval containing $\mu = 1{,}200$?

1,195 1,235 1,218 1,100 1,215

Answer: All the values, except 1,100 and 1,235, will result in a 95% confidence interval that contains $\mu = 1{,}200$.

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Section 8.4.1

Hot Spot #2 – Degrees Of Freedom

The t distribution requires that we find what are called degrees of freedom before we can find critical values for hypothesis testing. The letters df are used to represent the degrees of freedom. In both the t and chi-square distributions the degrees of freedom value is equal to one less than the sample size.

$$df = n - 1$$

The formula makes it simple to find the degrees of freedom, but it does not provide us any insight into what the phrase means. The idea behind degrees of freedom is illustrated by the following example:

- We want 3 numbers that have a sum of 10.
- We are free to choose any two numbers we want. For example we could choose 2 and 3.
- The third number must be a number that provides a sum of 10. In our example, the third number must be a 5. In this sense we do not have any freedom relative to the third number.

In our example we only had two degrees of freedom,

$$df = 3 - 1 = 2$$

If we had started with five numbers that had a sum of 10, we would have been free to choose any four numbers. However, the fifth number would not be a free number. Once we select four numbers, the fifth number is determined by the sum. The degrees of freedom would be $5 - 1$.

In the same way, if we started with n numbers that have a sum of 10, we are only free to choose $n - 1$ numbers. The last number is again determined by the sum; so, degrees of freedom equals $n - 1$.

The important point about the above examples is the effect that the sum 10 had on the numbers we selected. We used a sum of 10 to illustrate what happens, but any sum could have been used. For that matter we could have used the mean rather than a sum. The restriction that the numbers have a particular mean has the same effect of reducing our degrees of freedom to $n-1$.

In fact we have a similar loss of freedom for any set of numbers when the numbers are used in a calculation that must have a given value. It does not matter what calculation is done. The effect is the same. The sum was used in the above example because it is an easy calculation to do. The mean was used as a second example of a calculation that reduced the degrees of freedom because it involves a sum. However, we would have the same loss in degrees of freedom if we used the restriction that the numbers we select have a particular standard deviation.

One final point needs to be made about degrees of freedom that will be used in later chapters. The point involves what happens when we select numbers if more than one calculation is involved as a restriction. For example, if the numbers we select must be used in two calculations, we lose two degrees of freedom and write $df = n-2$. We can illustrate the idea using a familiar example from beginning algebra to show how we would lose two degrees of freedom. The problem is usually done with a variable and an equation that is then solved. In this case the problem is easy enough to solve that we will just give the solution by using observation.

- We want two numbers that have a sum of 10 and a product of 24.
- There are only two numbers that meet the requirement. The numbers are 6 and 4. We do not have any freedom. We started with two numbers, but lost two degrees of freedom because the numbers had to have a particular sum and product. Therefore, $df = 2-2 = 0$.

The above idea can be generalized in the following way:

> If the numbers we select must be used in k calculations that restrict the numbers we select, we lose k degrees of freedom and write $df = n - k$.

For example, if we had a large sample that we divided into k groups and required that the mean of each group be a specific value, we then would have $n - k$ degrees of freedom.

CHAPTER 8 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between a point and an interval estimate?

2. What is the error of the estimate?

3. How do you reduce the error of the estimate for μ?

4. What is meant by degrees of freedom?

5. How do you decide on the value of the z or t score to use with the confidence interval?

6. How does a t value compare to a z value?

7. How do we narrow the width of the confidence interval?

8. How is the Central Limit Theorem used in constructing confidence intervals?

9. When do we use s to find the upper and lower limits of the confidence interval?

10. Why do we use $p = q = 0.5$ when we find the sample size for a given error when estimating the population proportion?

CHAPTER 8 TEST

1. Given $\bar{x} = 1{,}225$, $s = 35$ and $n = 25$, find the point estimate of μ.

2. Given $\bar{x} = 1{,}225$, $s = 35$ and $n = 25$, find the 95% confidence interval for μ.

3. What is the t value for a 95% confidence interval with n=20?

4. If you want a smaller confidence interval for μ, what two choices do you have?

5. Given a sample of 50 scores with $\bar{x} = 1{,}237$ and $s = 125$, find the the value of E, the maximum error of the estimate, for the 95% confidence interval for the population mean.

6. What sample size is necessary to estimate the population proportion within 0.01 with 95% confidence?

7. A random survey of 100 managers found that 35 of them favored a national advertising campaign, find a 95% confidence interval for the population proportion, p.

8. What values of p and q should be used to find the sample size for a given value of E?

9. What is the value of α for a 95% confidence interval?

10. Which of the following confidence levels will give the narrowest interval: 90% or 95%?